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Supply chain collaboration in the presence of disruptions: a literature review

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The supply chain collaboration has gained significant attention, especially in the presence of disruptions. This paper presents findings from a systematic literature review to answer the question: how collaborations help supply chains respond and recover from a disruption. A total of 157 papers from the year 2000 were studied. The thematic and descriptive analysis identified usefulness, used research methodologies, collaboration mechanisms, and influential factors in collaboration. This comprehensive review provides in-depth insights into the current state of literature, proposes a research framework, and identifies several future research directions. It also highlights the role of each collaboration mechanism based on each severity level of disruptions.

Keywords: Supply chain resilience; disruption management; supply chain collaboration

Introduction

Recent disruption events remind us that we live in an unpredictable world. Disruptions not only cause inconvenience for organisations expecting to deliver or receive goods but also significantly reduce the financial performance of businesses along the supply chain (Ivanov et al. 2017). As an example, due to wildfires in Greece, flights to Athens airport had been diverted due to low visibility (BBC 2018). Alternatively, because of the disruption at Gatwick airport, EasyJet, the British airline, lost 15 million pounds as 400 flights had been cancelled, and 82,000 customers had been affected (CNBC 2019). As supply chains have been expanding globally, the exposure to possible disruptions has increased (Scheibe and Blackhurst 2018). The growing numbers of studies focussing on supply chain disruptions over the same period are evidence for the increased exposure (Tang and Musa 2011). This increasing trend is partly due to the widespread geographic human presence, transportation systems, and production facilities that are vulnerable to disasters (Wagner and Thakur-Weigold 2018).

Recent works (e.g. Ambulkar, Blackhurst, and Grawe 2015; Dolgui, Ivanov, and Rozhkov forthcoming; Dolgui, Ivanov, and Sokolov 2018; Gunasekaran, Subramanian, and Rahman 2015; Gupta et al. 2016; Ho et al. 2015; Ivanov et al. 2017; Ivanov, Pavlov, et al. 2016; Ivanov, Sokolov, et al. 2016; Ivanov and Dolgui 2019; Khalili, Jolai, and Torabi 2017; Namdar et al. 2018; Scheibe and Blackhurst 2018; Tukamuhabwa et al. 2015; Yu, Li, and Yang 2017) extensively investigated reasons for disruptions and their impacts on the supply chain performance. We invite readers to visit recent reviews of supply chain disruption (e.g. Ivanov et al. 2017; Shen and Li 2017; Snyder et al. 2016) for a comprehensive background.

When having a disruption, organisations are even more vulnerable due to a lean, global, and flexible supply chain (Pettit, Croxton, and Fiksel 2013). Hence, supply chain managers aim to reduce the negative effects of disruptions by providing proactive strategies that increase the awareness of organisations (Mohammaddust et al. 2017). This expectation requires the abilities to explore the whole supply chain network to respond and recover after disruptions. Subsequently, the collaboration has been highlighted as an important activity in building resilience (Pettit, Croxton, and Fiksel 2013). Collaboration is the abilities of organisations to work together, to plan, and to execute supply chain operations toward common targets (Cao et al. 2010).

Despite the consensus on the importance of collaboration, it remains a gap in the literature on understanding how exactly collaboration helps supply chains respond and recover from disruptions. First, it is unclear how collaboration relates to the abilities in building resilience. Christopher and Peck (2004) recognised collaboration as one of the constructs of resilience. By definition, supply chain vulnerability is a network-wide concept (Soni, Jain, and Kumar 2014). A high level of collaboration across supply chains helps organisations significantly respond and recover from disruptions (Christopher and Peck 2004).

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In contrast, when treating collaborative communication and information-sharing separately from joint practices, Scholten and Schilder (2015) identified collaboration as an antecedent of constructs of resilience (i.e. visibility, velocity and flexibility). Especially, Scholten and Schilder (2015) found that decision synchronisation with good quality, reliable, and complete information enables readiness and quick response to disruptions. Considering literature systematically, we aim at providing a comprehensive picture of the relationship between collaboration and supply chain resilience.

Second, in the presence of disruptions, how supply chain partners collaborate, and what collaborative activities should be prioritised remains unclear. While Pettit, Croxton, and Fiksel (2013) suggested that single sourcing increases the effects of disruption, Ergun et al. (2010) proposed that focusing on single sourcing can make response processes faster. Third, although collaboration with upstream and downstream partners are mostly discussed in the literature, Chen et al. (2017) stated that collaboration with competitors and other organisations at the same level need to be considered.

Given these issues, the primary objective of this research is to investigate insights into the role of collaboration in the presence of disruptions that leads to resilient supply chains. This research addresses the call by Natarajarathinam, Capar, and Narayanan (2009) to have more studies in investigating responses and recoveries from disruptions. It is in line with suggestions of Soosay and Hyland (2015) who advocated that future research should consider the environmental uncertainties and the circumstances where the collaboration happens. It also closes to the review on disruption recovery of Ivanov et al. (2017), who restricted to quantitative papers and left collaboration strategies as a future research direction.

In so doing, this research makes three contributions to the supply chain literature. First, this research is one of the first works to analyse the role of collaboration between organisations when dealing with disruptions. With insights into not only a focal organisation but also suppliers, buyers and their interactions, this research investigates the resilience at the whole supply chain level. Supply chain collaboration relates to activities such as information or resources sharing at different organisations, which lead to more visibility, velocity, and flexibility in the supply chain (Cao et al. 2010). When organisations share information, they also share knowledge of processes and procedures that improves their resilient capabilities. Second, this research provides further light on the importance of collaboration in supply chain resilience. It provides detail of collaborative activities that cause the supply chain resilience and the interaction of these activities.

Third, this research also aims to provide guidance for managers on how to use collaborative activities to respond and recover from disruptions. For example, in practices, organisations may delay sharing information when having disruptions. This helps organisations to resolve issues internally (Pettit, Croxton, and Fiksel 2013) but may reduce resilient capabilities for the whole supply chain. This research will provide clear guidance for managers in such scenarios to improve organisations' capabilities when dealing disruptions.

The rest of the research is structured as follow. The next section clarifies key relating concepts and definitions. We then present the research method and the sampling process. Subsequently, we discuss descriptive analysis results and findings from the thematic synthesis approach. Finally, we propose a research framework, suggest several future research directions, and conclude the paper.

Literature review

This research bases on three theoretical foundations, including supply chain resilience, supply chain sustainability, and collaboration. The following sections review relevant literature on these three foundations.

Supply chain resilience

Disruptions in supply chains relate to any disturbance to the flow of materials, products, or services. As disruptions are unavoidable, organisations mitigate the effects of disruptions by adopting traditional supply chain practices but cannot prevent all of them. Therefore, investigating approaches to build capabilities for supply chains to deal with disruptions is an interest for both academics and practitioners (Scholten and Schilder 2015). Resilience provides supply chains capabilities to prepare for disruptions, to reduce the impact of disruptions, and to recover quickly from disruptions (Christopher and Peck 2004).

As supply chain resilience is a network-wide concept, all partners within the supply chains have to adopt the formative resilient elements to respond and recover from disruptions. Thus, collaboration is important in supply chain resilience as it supports higher service level, visibility and flexibility level, and lower cycle times (Cao et al. 2010). Despite these findings, it is unclear how exactly collaboration affects supply chain resilience. Awareness to understand the relationship among resilience factors, response and recovery from disruptions remains a gap in the literature (Scheibe and Blackhurst 2018).

Response and recovery phases

In the disaster management cycle, the operations of an organisation can be classified into four phases, including preparedness, mitigation, response, and recovery (Altay and Ramirez 2010). In general, preparedness and mitigation phases refer to activities that are performed at the pre-disaster phase to prepare for possible disasters and to prevent disaster occurrence. The response and recovery phases refer to post-disaster activities which aim to manage limited resources efficiently and effectively when responding to damages caused by disasters. While most of the literature look into preparedness and mitigation phases, not many studies look into response and recovery phases that specify plans for getting back to normal operations (Natarajarathinam, Capar, and Narayanan 2009). This trend displays challenges and calls for more research on the response and recovery phases.

Supply chain sustainability

In response to increasing disruption threats caused by disasters, there are continuous efforts to propose an efficient supply chain that is able to jump back from any disruption and is sufficient to pose the same sustainability under disruptions. These efforts lead to growing attention to resilience and sustainability. In general, sustainability considers environmental, social, and economic aspects of the present and future generations, while resilience focuses on the response of systems to disturbance and persistent stress (Dolgui, Ivanov, and Sokolov 2018).

The majority of literature studies investigate sustainability and resilience concepts independently. However, in reality, there are many situations where sustainability practices provide capabilities for supply chains to tackle disruptions. For example, increasing efficiency (e.g. less safety stock levels) is a common practice in sustainability. While this practice improves the environmental and economic aspects of an organisation, it also limits the availability of inventory to deal with supply and demand variations. Therefore, it is more realistic to consider resilience when performing sustainable supply chain management and vice versa (Ivanov 2018).

Collaboration

The supply chain is an integration of business processes or organisations; thus, to properly manage the impact of disruptions on supply chains, we need to understand this integration. In this context, collaboration is a strategic response to issues happening in the chain (Altay and Ramirez 2010). Cao et al. (2010) considered collaboration, including activities of information sharing, resources sharing, communication, joint knowledge creation, joint decision-making, and targets congruence among all supply chain partners.

It is noted that there are differences in terms of collaboration and coordination. While coordination refers to resource sharing or activities that make the consumer demand information available to the whole supply chain, collaboration goes beyond mere information exchange and resource sharing (Moshtari 2016). However, existing literature has often used coordination and collaboration interchangeably (Balcik et al. 2010; Dubey et al. 2019). Thus, this research considers papers discussing both collaboration and coordination for generating broad insights from the literature.

There are two main perspectives of supply chain collaboration in the presence of disruptions including commercial and humanitarian supply chain (Balcik et al. 2010; Jahre and Jensen 2010; van Wassenhove 2006). The significant differences between commercial and humanitarian are demand prediction and consequences of disruptions (McLachlin and Larson 2011). The commercial supply chain has advantages of historical demand, whereas anticipating a disaster time and location is almost impossible. Beyond that difficulty, a failure in operating humanitarian supply chains may lead to suffering or loss of life (Day et al. 2012). Moreover, while cost efficiency is a key criterion in the commercial supply chain, a humanitarian supply chain focuses mostly on the reduction of lead times rather than costs, at least for a few hours after the disruption (McLachlin and Larson 2011). Other issues that could lead to different responsive actions under the humanitarian supply chain, including authority, socio-economic, policy, logistical, or financial issues (Day 2014). This section aims to review various collaborative efforts in responding to disruptions in the commercial and humanitarian settings.

Commercial supply chain

Under a disruption, organisations need strategies that promote the recovery to stabilise and adapt their supply chains in order to ensure continuous performance and avoid long-term impacts. Altay and Ramirez (2010) analysed over 3500 disasters and indicated that responsive actions should be supply chain-wide rather than company-specific practices. Brusset and Teller (2017) found that collaborative activities such as allowing continuous inventory adjustment and sharing forecast and demand result in a high level of supply chain resilience. Ivanov, Sokolov, and Dolgui (2014) proposed using the ripple effect to consider the propagation of a disruption downstream the supply chain. Strategies such as collaborative planning, forecasting and replenishment (Dolgui, Ivanov, and Rozhkov forthcoming; Dolgui, Ivanov, and Sokolov 2018), resources allocation (Ivanov, Pavlov, et al. 2016), coordinated production-ordering contingency policy (Ivanov and Sokolov forthcoming), and identification of performance measurements (Hosseini and Ivanov forthcoming) make supply chain more flexible and reduce the effect of disruptions. For these reasons, it is necessary to understand how to collaborate and to identify important factors that influence collaborative activities (Ivanov and Dolgui 2019).

Humanitarian supply chain

The application of collaboration in humanitarian supply chains has been received increased attention recently (Day et al. 2012; Fawcett et al. 2015; van Wassenhove 2006). Collaboration helps the supply of humanitarian aids timely and improves overall response efficiency (Naor et al. 2018). Collaboration activities through exchanging knowledge, information, and resources help humanitarian organisations effectively respond to disruptions (Moshtari 2016; Wagner and Thakur-Weigold 2018). It is noted that information and communication technologies have a vital role in collaboration among humanitarian organisations (Ergun et al. 2014). With the rapidly growing of big data, understanding of how big data analytics enables trust and collaboration is a promising research area (Dubey et al. 2018; Dubey et al. 2019).

Despite these benefits, researchers and practitioners have underlined the lack of humanitarian collaboration, due to (i) the desire to attract media attention, (ii) differences in organisational strategies, structures, and interests, (iii) lack of respect and trust, (iv) operational conditions, or (v) high level of supply and demand uncertainties (Balcik et al. 2010; Eftekhar et al. 2017; Ergun et al. 2014; McLachlin and Larson 2011; van Wassenhove 2006). A lack of collaboration could result in ineffective responses and major losses for disaster-affected areas. After a disaster, organisations need to focus on recovery, rehabilitation, and reconstruction. During this stage, besides transparency and accountability, the relationship among organisations is important (Altay et al. 2018). Thus, collaboration has been considered as a critical factor for success in the humanitarian supply chain (Gunasekaran et al. 2018).

In the humanitarian supply chain, many factors contribute to successful collaboration, including commitment, cultures, trust, information sharing, and mutual respects (Fawcett et al. 2015; Prasanna and Haavisto 2018). When a disaster happens, organisations often collaborate with the military to implement humanitarian programmes (Apte, Gonçalves, and Yoho 2016). Naor et al. (2018) argued that the global need for humanitarian activities and collaboration among civil–military partners is on the rise. These studies highlighted the importance of civil–military collaboration in humanitarian supply chain and called for studies that shed lights on how the collaboration works in the humanitarian setting.

Research methodology

We conducted a systematic literature review, which produces reliable knowledge and enhances practices (Tranfield, Denyer, and Smart 2003). A systematic literature review method suits well to this research as it is extensive regarding the range of contexts, businesses, and operations policies. Extensive ranges help identify an in-depth understanding of how organisations collaborate to deal with disruptions and generate insights into contexts and processes that are lacking in the current literature. Accordingly, we followed the procedure proposed by Tranfield, Denyer, and Smart (2003) (Figure 1). Although this method is time-consuming and labour intensive, it provides high quality and most efficient results in many comprehensive literature review works (e.g. Carter and Easton 2011; Hietschold, Reinhardt, and Gurtner 2014; Marodin and Saurin 2013; Soosay and Hyland 2015).

Data identification

This process includes selecting relevant keywords and data sources. It is an iterative and necessary to evaluate the importance of literature (Tranfield, Denyer, and Smart 2003). We initiated the process by examining recent works on supply chain resilience and collaboration (e.g. Chen et al. 2017; Dolgui, Ivanov, and Rozhkov forthcoming; Dolgui, Ivanov, and Sokolov 2018; Ivanov et al. 2017; Ivanov and Dolgui 2019; Scholten and Schilder 2015). Through this examination, we identified keywords and constructed search terms around combinations of keywords relating to the supply chain, collaboration, and disruption.

The first set of keywords (supply chain, supply) relates to supply chain management. The second set of keywords (coordinate*, collaborate*, cooperate*, partnership) relates to collaboration. The third set of keywords (sustainable*, risk*, resilience*, robust*, redundancy*, recovery*, response*, relief*, adapt*, disrupt*, disaster*) relates to disruptions. While we are aware of the distinction between sustainability and resilience, we still included sustainability keyword in our search to have a good range of articles. These keywords were also based on comments from scholars at a conference in September 2019.

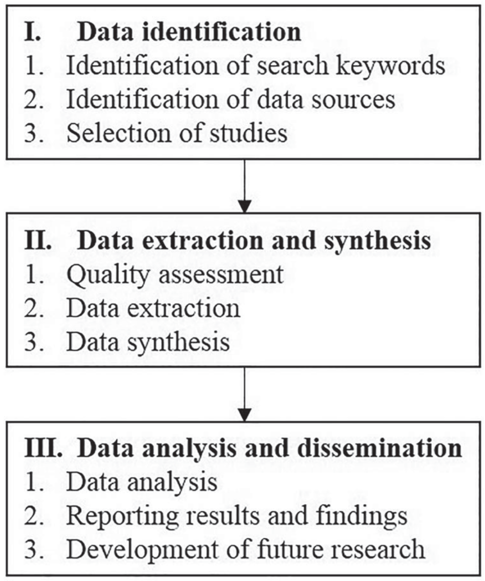

Figure 1. Systematic literature review procedure (Adapted from Tranfield, Denyer, and Smart 2003).

The papers were retrieved from the Scopus and Web of Science library databases from the year 2000 to have good coverage and to have emerging works on the supply chain resilience (Snyder et al. 2016). We refined the search by keeping only documents written in English. There are 4917 and 4123 articles extracted from Scopus and Web of Science databases. In total, after removing duplicated records, we have 4992 articles.

Data extraction and synthesis

In common with Soosay and Hyland (2015), we focused on high-quality journals that are included in the ABS Academic Journal Guide 2018, or the 2019 Australian Business Deans' Council (ABDC) lists, or having an impact factor, or 2018 SCImago journal rank over 1. After this stage, 1443 papers were retained.

Thereafter, we thoroughly inspected each article's title and abstract to evaluate whether the article discusses supply chain collaboration in the context of disruption. We also excluded articles focusing on physical and health science areas such as 'chemical engineering', 'medicine', 'immunology', biochemistry', and 'chemistry'. This resulted in 194 articles for full-text assessment. Although 37 articles contained selected keywords, they were not relevant for the study, such as they focused on the preparedness phase or did not discussed the supply chain collaboration. This concluded the final list of 157 articles for data synthesis and analysis. Figure 2 presents an adopted PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) approach (Moher et al. 2015) followed for screening the dataset.

Data analysis and dissemination

Data analysis includes descriptive and thematic analysis. While descriptive analysis provides an overview of selected articles, thematic analysis is more comprehensive and interpretative (Cloutier, Oktaei, and Lehoux forthcoming). To enhance the quality of this review work, we used NVivo 11, which is considered a powerful software for managing, visualising, and analysing qualitative data (Jackson and Bazeley 2019). The 157 selected articles were imported from a reference management software (Zotero) to NVivo and identified in the (author, date) format.

Table 1 shows the most important words in the sample by using the 'Word frequency' function in NVivo 11. Most of the words in Table 1 show a strong correlation with search keywords and thus, validate the identified keywords. It provides confidence in the reliability of the following analysis.

Descriptive analysis

This section provides generically descriptive findings from the sample. Figure 3 presents the trend in the number of articles published from 2000. There is an increasing trend over the years, especially from 2015. This trend represents the growing interest in supply chain collaboration and disruptions (Soosay and Hyland 2015). The largest number of articles

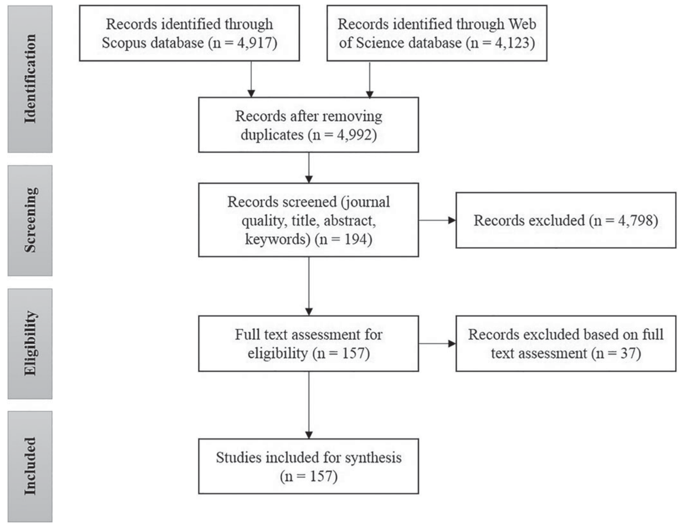

Figure 2. PRISMA flowchart for data screening.

Table 1. The most frequent keywords through text mining.

Word	Frequency	Word	Frequency	Word	Frequency
supplying	20988	quantity	3735	problem	2161
chain	12473	decisive	3564	provide	2157
coordinate	8585	humanitarian	3530	distributions	2076
retailing	8279	logistics	3410	functions	2035
products	7827	timing	3231	valuing	1977
managing	6988	sharing	2966	expects	1968
demand	6420	increasingly	2859	develops	1907
disrupts	6389	international	2858	organizing	1896
model	5975	effect	2711	resources	1893
costs	5515	responsiveness	2690	impacts	1880
pricing	5477	reliefs	2548	emerging	1856
contracts	5413	markets	2514	channel	1854
operators	5122	follows	2431	servicing	1822
manufacture	5076	inventory	2430	plans	1817
risks	5017	differs	2412	policy	1783
order	4723	levels	2389	resilient	1771
optimizing	4457	performs	2307	networks	1734
profits	4184	informs	2301	revenues	1709
disaster	4159	capacity	2273	uncertainty	1704
systems	3819	strategy	2175	process	1663

was observed in 2018. While the number of articles in 2019 decreased, it might be because the sample was collected in September 2019 and thus, did not cover the whole year.

The selected articles were published in 40 journals. Figure 4 shows the top ten journals most frequently published, namely, *International Journal of Production Research, International Journal of Production Economics, European Journal of Operational Research, Computers & Industrial Engineering, Journal of Humanitarian Logistics and Supply Chain Management, Annals of Operations Research, Omega, International Journal of Disaster Risk Reduction, Journal of Cleaner Production, and Production Planning & Control*. These journals account for 67 per cent of sample and have been dedicated to publishing research on disruption, disaster management, and supply chain collaboration. The remaining articles were published on 30 interdisciplinary journals, including management science, technology management, or general management areas.

The thematic analysis was conducted to deliver a systematic literature review of how supply chain partners collaborate to respond and recover from a disruption. Each article was read carefully to identify key themes for analysis, such as collaboration mechanisms or usefulness of collaboration. During this iterative process of screening and synthesis, other themes were identified as follows.

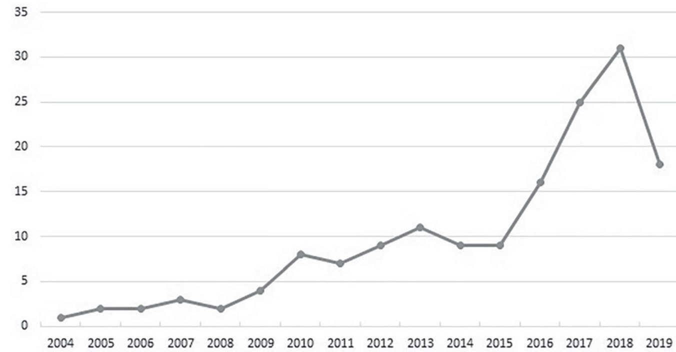

Figure 3. Total number of articles published by year.

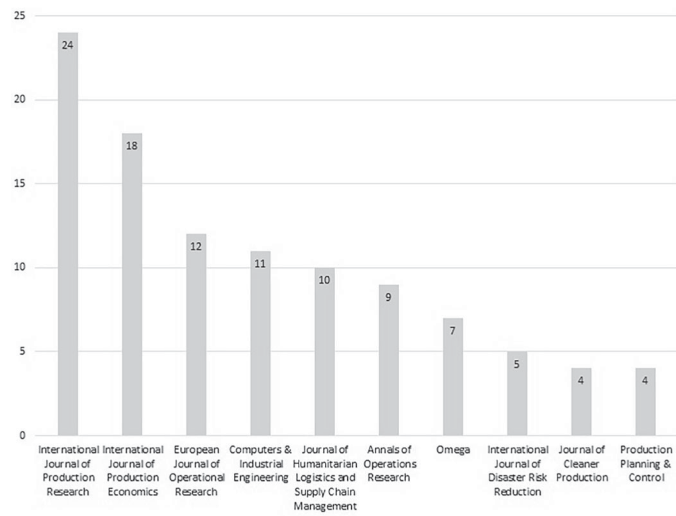

Figure 4. Top ten journals most frequently published.

- *Collaboration mechanisms* that supply chain partners adopt to collaborate to cope with disruptions.
- *Reactions* to disruption events.
- *Influential factors* in collaboration.
- *Usefulness* of collaboration in the presence of disruptions.
- *Research methodologies* that are commonly used in the literature.
- *Disruption sides* or the sources of disruptions and focuses in the literature.
- *Types of a supply chain* include commercial and humanitarian supply chains.
- *The number of echelons* that identifies the complexity of the studied models.

Research methodologies

We broadly categorised research methodologies into quantitative, qualitative, and mixed methods. Quantitative methodology introduces papers that applied mathematical, optimisation, simulation modelling, or statistical analysis. Qualitative methodology includes surveys or interviews, conceptual framework development, case studies, or literature reviews. Mixed methods consider both quantitative and qualitative method. Table 2 summarises research methodologies employed in the sample. Mathematical modelling was the preferred approach for commercial and humanitarian supply chains. Most of these papers studied supply chain contracts under a disruption (Cai et al. 2019), advantages of back-up suppliers (Chakraborty, Chauhan, and Ouhimmou forthcoming), or identification of the optimal location (Kamyabniya et al. 2018). Simulation method was used to identify the scheduling process of emergency supplies (Jia and Kefan 2015) or to assess the system performance under different scenarios (Li et al. 2019). Statistical methods were used to characterise the relationship between explanatory and dependent variables (Eftekhar et al. 2017).

Table 2. Research methodologies in the sample.

Research methodology		Commercial	Humanitarian
Mixed methods		2	3
Qualitative methods	Case studies	2	9
	Conceptual paper	2	3
	Interview-survey	4	10
	Literature review	3	4
Quantitative methods	Mathematical modelling	86	13
	Simulation	7	5
	Statistical analysis	1	3

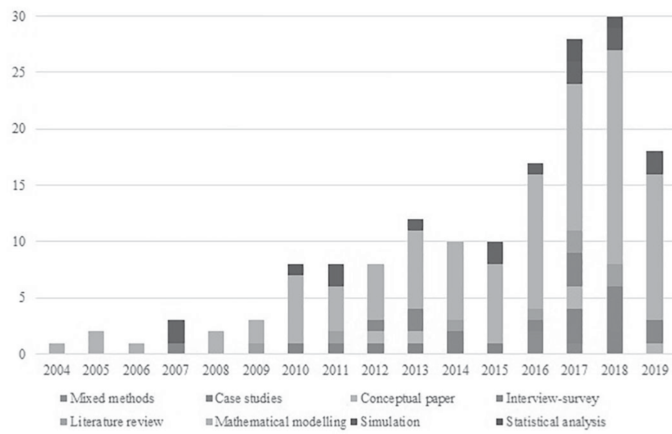

Figure 5. Research methodologies by year.

In qualitative methods, most of the papers related to humanitarian supply chains. It is noted that interviews, surveys, and case studies were favoured methods. These methods were applied to explore core capabilities of the military and non-military organisations involved in humanitarian operations (Apte, Gonçalves, and Yoho 2016) or the potentials of outsourcing of humanitarian logistics activities to logistics service providers (Sigala and Wakolbinger 2019).

Figure 5 shows that mathematical modelling methods have been used dominantly in every year since 2004. There was also an increasing trend of using interviews and surveys methods since 2015. While simulation has been used frequently, it is not a dominant method.

The number of echelons

This theme attempts to capture the complexity of studied models. The majority of the research (83.5%) studied two-echelon models, with limited studies focusing on three-echelon (12.6%) and four- or five-echelon models (3.9%). Two-echelon research included one supplier – one buyer (Cai et al. 2017), one supplier – buyers (Chen and Zhuang 2011), and suppliers – buyers (Adem et al. 2018). Three-echelon research included one supplier – one manufacturer – one buyer (He and Zhao 2012; Zhu, Krikke, and Caniëls 2016) and suppliers – manufacturers – buyers (Ivanov, Pavlov, et al. 2016; Wakolbinger and Cruz 2011). Limited research considered four-echelon models (Altay and Ramirez 2010; Song, Chen, and Lei 2018).

Disruption sides

With respect to disruption sides, 39.5% of papers referred to the demand disruption, including response to natural disasters (Akhtar, Marr, and Garnevska 2012; Nadi and Edrisi 2017), products with long lead times and short selling season (Arikan and Silbermayr 2018), new launched products (Buzacott and Peng 2012), and lack of updated demand information (Song, Chen, and Lei 2018). Network-based papers (35.1%), considering supply and demand disruptions simultaneously, examined the network resilience (Beheshtian et al. 2017), and the presence of demand and production disruptions (Giri and Bardhan 2015). The rest, (25.4%) supply-based disruption papers, included responses to yield uncertainties (He and Zhao 2012) and responses to unreliable suppliers (Huang, He, and Li 2018). It is noted that the inaccurate demand information drives the demand side while the yield uncertainty dominates the supply side.

Types of supply chains

In term of types of supply chains, 69.5% of papers covered commercial supply chains. These papers covered a wide range of topics such as transportation model (Paul et al. 2019), optimal pricing and production scheduling (Yao, Xu, and Luan 2016), or behaviour of decision-makers (Wakolbinger and Cruz 2011). 30.5% of papers dealt with humanitarian supply chains such as logistics service providers (Baharmand, Comes, and Lauras 2017; Bealt, Barrera, and Mansouri 2016), the use of big data (Dubey et al. 2018; Dubey et al. 2019), or locations of relief centres (Wilson et al. 2018). The predominance of commercial supply chains in our sample is in line with previous studies (Balcik et al. 2010; Ergun et al. 2014; Prasanna and Haavisto 2018), which suggest that collaboration in the humanitarian supply chain is still in the immaturity stage and could apply many collaboration practices from commercial supply chains.

Thematic analysis

This thematic analysis section first identifies collaboration mechanisms to cope with disruptions, followed by the usefulness of collaboration, influential factors in collaboration, and collaborative reactions to disruptions.

Collaboration mechanisms

There was a vast of collaboration mechanisms proposed in the sample; it proved that supply chains could collaborate through many methods. To group these mechanisms, we followed seven categories proposed by Cloutier, Oktaei, and Lehoux (forthcoming), including contractual and economics practices; joint practices; relationship management; technological and information sharing practices; governance practices; assessment practices; and supply chain design. These categories were developed in sustainable supply chain development and therefore seemed pertinent for this research. Each category is presented with examples of specific mechanisms and their roles to cope with disruptions as follows.

Contractual and economics practices. The contract is a common form to collaborate with supply chain partners (Cachon 2003). A contract is designed with legal agreement to increase the total profit and to reduce risks among supply chain partners. A contract specifies terms and must be enforceable, i.e. supply chain partners must fulfil their promises. Thus, the terms mentioned in the contract should be verified (Coltman et al. 2009). This category also includes economics practices such as profit-sharing, rewards, and incentive alignments. The mechanisms in this category identify expectation from supply chains and support collaboration among partners.

A key finding from studied articles that sheds light on the collaboration mechanism is the dominant of the revenue-sharing contract (Cai et al. 2019). Under this contract, besides a wholesale price for each purchased unit, a retailer pays a supplier a percentage of revenue generated by the retailer; thus this contract is beneficial to all partners in a supply chain (Cachon 2003). In the reverse supply chain, a revenue-sharing contract could be used to convince the retailer's decision on reward amount and jointly optimise customers' willingness to return (Heydari, Govindan, and Sadeghi 2018). The less common contract form is the wholesale price (Chakraborty, Chauhan, and Ouhimmou forthcoming). As this contract requires buyers to pay a fixed price and quantity to suppliers, the suppliers have to adjust price and quantity according to demand disruptions. This contract is beneficial to utilise emergency suppliers and guarantee supply chain collaboration (Inderfurth and Clemens 2014).

Incentive alignment has also been employed for responses over disruptions. It is the process that supply chain partners are willing to share costs, benefits, and risks so that the collaboration could be equally realised (Vakharia and Wang 2014). When a supplier experiences a disruption, it may face difficulties in resuming production, causing a shortage for the manufacturer and the whole supply chain. Chen, Shen, and Feng (2014) indicated that the manufacturer must provide incentives to retailers to maintain the order quantity. Sigala and Wakolbinger (2019) explored the potential of using incentives for the outsourcing of humanitarian logistics to commercial logistics service providers.

Joint practices. This category addresses mutual reactions among supply chain partners from joint planning to joint control activities (Mentzer et al. 2001). It facilitates collaboration as it establishes consensual plans between partners. Dolgui, Ivanov, and Rozhkov (forthcoming) proposed a contingent production-inventory control policy to mitigate the ripple effect as a result of the disruption. Another example, given by Shahparvari and Bodaghi (2018), was about organisations wishing to collaborate on reducing the time window for relief supply. These authors proposed an integrated decision model for humanitarian operations to accomplish this effort.

In addition, we observed that supply chain partners arranged their decisions with each other, called decision synchronisation, to maximise the response and recovery processes. The importance of such synchronised decisions in demand management, scheduling, or distribution management has been highlighted (Scholten and Schilder 2015). Under a disruption, the market demand highly fluctuates and leads to the high variation of costs. Thus, the centralised decision could provide guidance for retailers to set the retail price and the replenishment plan that maximise the profit of the whole supply chain (Chen and Zhuang 2011).

Supply chain design category considers the responsiveness structure for the supply chain. This category facilitates collaboration by proposing decision-making tools for supply chain partners. For example, Hlioui, Gharbi, and Hajji (2017) proposed an integrated production, replenishment, and raw material quality control policy to help with supplier selection. In the humanitarian setting, dynamics models have been developed for deciding locations of relief centres (Wilson et al. 2018) or choosing a transportation model (Apte, Gonçalves, and Yoho 2016; Mansoornejad, Pistikopoulos, and Stuart 2013).

Relationship management category facilitates collaboration through activities such as communication, motivation, or training to enhance the relationship among stakeholders. Once a disruption happens, it could rapidly affect the whole supply chain. Thus, this category is important to raise awareness of the issues as well as seeking commitment from stakeholders for coping with the disruption. For example, Nurmala, de Vries, and de Leeuw (2018) showed that humanitarian-business collaboration facilitates resources, knowledge, and skills and improves the effectiveness and efficiency of humanitarian logistics. Other papers identify characteristics for building good relationships such as trust (Day et al. 2012; Li et al. 2019), communication (Wagner and Thakur-Weigold 2018), and training (Eftekhar et al. 2017).

Assessment practices. One of the problems on the response phase is that the effects of disruption are over or underestimated. Thus, assessment should be carried out to provide accurate information on the needs after a disruption. Assessment is the second core capability that humanitarian organisations must be cultivated (Apte, Gonçalves, and Yoho 2016). Nadi and Edrisi (2017) developed a collaborative demand evaluation model, which improves the humanitarian assessment after a natural disaster and emergency response operations. These practices help supply chain partners provide feedback, make better decisions, and motivate for better performance (Akhtar, Marr, and Garnevska 2012). To this effect, information sharing has been highlighted as a key enabler for assessment practices (Apte, Gonçalves, and Yoho 2016).

Technological and information sharing practices introduce activities and data that provide relevant and accurate information for supply chain collaboration. The accuracy, relevance, and timeliness of information are necessary for situation awareness and to determine the plan for responses after a disruption. Brusset and Teller (2017) highlighted the advantages of enterprise resource planning (ERP) technology in humanitarian supply chains. Wakolbinger and Cruz (2011) proposed a framework that enables information sharing for the risk-sharing contract. They showed that innovations that reducing information sharing costs could lead to equal benefits for all supply chain partners.

Governance practices. This category addresses rules, laws, and policies that manage organisations, systems, and activities. For example, the government should monitor the transportation of emergency supplies on a daily basis (Jia and Kefan 2015) or embrace the public-private collaborations in managing the response and recovery phases of disruptions (Sigala and Wakolbinger 2019). There are negative and positive governance conditions (e.g. communication, strategy) that could shape a collaboration (Gabler, Richey, and Stewart 2017). Thus, supply chain partners should consider a variety of stakeholders who involve in the response and recovery phases (Heaslip, Sharif, and Althonayan 2012).

In summary, there are seven collaboration mechanisms for a supply chain collaboration in the presence of disruptions. Not surprisingly, while commercial supply chains have adopted all of these mechanisms, we found that humanitarian supply chains have not focused on the contractual and economic mechanism. It is reasonable as the nature of humanitarian supply chains is to provide quick emergency supplies. Thus, humanitarian supply chains have mostly focused on joint practices or relationship management.

Reactions to disruptions

Under a disruption, organisations often have limited resources, which require adaptation in business processes. In such scenarios, organisations cannot expect to apply the same level of priority to every aspect of the business. Instead, depending on the severity of disruptions, supply chains partners could consider parametrical adaptation, process adaptation, or structural adaptation (Ivanov et al. 2017).

Parametrical adaptation characterises the response and recovery through adjusting critical parameters such as production plan. After a disruption, there is a benefit zone where buyers and suppliers can benefit from the adjustment of contract (Zhang, Xiong, and Xiong 2015); the more intensive competition in the supply chains, the sooner these adjustments should be taken (Wu et al. 2018). Most studies shed light on the response at the upstream level as the supply chain collaboration is initiated mostly by the upstream level. The supplier could charge a higher wholesale price when having a positive demand disruption (Chen and Zhuang 2011). When the lead time increased, the supplier would lower the wholesale price to stimulate demand (Xiao and Jin 2011). On the downstream side, a disruption could affect the operations of retailers and the ability to bring products to consumers. Consequently, the role of retailers has been received much attention in designing a contract (Zhang, Xiong, and Xiong 2015). The circumstances in which the role and behaviour of retailers might develop, therefore require further investigation.

Other studies report on when the production plan should be modified. Changes to the original plan could cause ripple effects and bullwhip effects (Dolgui, Ivanov, and Sokolov 2018) and impose additional costs throughout the supply chain (Qi, Bard, and Yu 2004). Thus, if the disruption is small, the supplier and the buyer could keep the original supply and order plan (Zhang, Xiong, and Xiong 2015).

Process adaptation considers the flexibility of capacity. It is important to consider the supply chain network and the partnership with logistics service providers when following process adaptation. It is because when the production capacity increases, it might require new warehouses or transportation systems (Mansoornejad, Pistikopoulos, and Stuart 2013). Under capacity disruptions, investments on surge capacity at the supplier and logistics service providers are useful for recovery (Zhu, Krikke, and Caniëls 2016).

Structural adaptation considers modifications of network structure and deployments of assets (e.g. backup suppliers) (Ivanov et al. 2017). In the presence of disruption, it is preferable to take advantages of a backup supplier (Chakraborty, Chauhan, and Ouhimmou forthcoming). In this scenario, a portion of order quantity shifts to a backup supplier, thus, the primary supplier should encourage more order quantities from the retailer by reducing the wholesale price (Giri and Bardhan 2015). Contingency plans are also considered for response and recovery as relief supplies play an important role in minimising suffering. Alternative transportation plans and schedules have been studied in allocating relief resources and support as many victims as possible (Shahparvari and Bodaghi 2018).

Usefulness of collaboration

The collaboration creates usefulness such as better capabilities and resources, finance, and service improvements in responding and recovering after a disruption.

Capabilities and resources. The common purpose for supply chain partners to collaborate is sharing resources and capabilities such as communication infrastructure, financial, and non-financial resources. The collaboration enhances information and knowledge sharing, and thus, improves operational efficiency and overall profits (Gabler, Richey, and Stewart 2017). Effective allocations of financial resources (e.g. budget) and non-financial resources (e.g. medicine or food) are required to manage collaborative teams after disruptions (Li et al. 2019). It is noted that leadership has a critical role in the collaboration, especially for humanitarian supply chains where many organisations and people involved (Akhtar, Marr, and Garnevska 2012).

Financial benefits. It is difficult for organisations to control costs because of uncertainties after a disruption. Transportation could be the most crucial and difficult cost to manage (Paul et al. 2019) due to the increased demand and limited roads or vehicles. The inventory cost is challenging as organisations increase stock to cover demand variations. Thus, supply chain partners could share warehouses, vehicles, or labour to lessen financial losses (Baharmand, Comes, and Lauras 2017).

Service improvements. Collaborative activities could lead to shorter lead times or more accurate information; thus, service improvements. For example, the collaborative development of inventory policies allows reduction of backlogs and adjustment of inventory levels (Dolgui, Ivanov, and Rozhkov forthcoming). The collaboration could also improve transparency and accountability between partners (Heaslip, Sharif, and Althonayan 2012) and foster innovative solutions to challenges after disruption (Maon, Lindgreen, and Vanhamme 2009).

Influential factors in collaboration

Apart from usefulness, the thematic analysis also identifies the following factors that influence collaboration.

Information sharing and technology, described under different synonyms, including information exchange, communication, and risk sharing, are the foundation of collaboration (Cao et al. 2010). They enable learning from previous disruption events, facilitate risk assessment, and evaluate the value of relationship (Chen et al. 2017). This factor is critical for collaboration and risk reduction (Christopher and Peck 2004) and effects on the system's abilities to deal with disruptions (Soni, Jain, and Kumar 2014). It helps monitor performance and identifies potential problems in supply chains (Macdonald et al. 2018).

Trust is fundamental for collaboration (Dubey et al. 2019); one should believe that their partners are willing and able to accomplish their duties. Once a disruption happens, there is a potential breach of collaboration, which might provide some feelings that the partners are untrustworthy. A high level of trust leads to the better collaborative relationship (Prasanna and Haavisto 2018) and higher supply chain performance in the commercial and humanitarian settings (Eftekhar et al. 2017; Li et al. 2019).

Culture refers to beliefs and shared values that explain for practices in the supply chain (Prasanna and Haavisto 2018). Organisations with a collaborative culture tend to collaborate with other partners actively. Different cultures lead to collaborative failure between buyers and suppliers (Fawcett et al. 2015) or hamper effective relief responses (Prasanna and Haavisto 2018).

Stakeholders refer to support and commitments from partners for the collaboration (Altay et al. 2018). A strong relationship with stakeholders could cover all the basic needs during a disruption (van Wassenhove 2006). The collaboration may also happen due to the pressure from stakeholders (Heaslip, Sharif, and Althonayan 2012). Therefore, establishing with the right partner is the priority to secure response activities. Furthermore, the government is one of the key stakeholders, who could strive to collaborate and notice the lack of collaboration (Akhtar, Marr, and Garnevska 2012).

Divergent goals describe inconsistent and disconnected goals between partners. These are barriers to supply chain collaboration as partners' priorities may not align (Gabler, Richey, and Stewart 2017). For a successful collaboration, attention should be focused on priorities of objectives and the nature of the supply chains (Paul et al. 2019).

Flexibility indicates partners' willingness to adapt and change for collaboration purposes. New strategies of flexibilities, such as responsive and agile supply chains (Dolgui, Ivanov, and Sokolov 2018) help organisations to focus on reactive capabilities (Altay et al. 2018).

Knowledge and experience are other barriers for the collaboration, especially when a disruption happens at a new region (Adem et al. 2018). Lack of knowledge and experience is associated with behavioural barriers (Bealt, Barrera, and Mansouri 2016). Thus, comprehensive solutions such as advisory and training are necessary for the success of collaboration (Naor et al. 2018).

Market factor covers requirements from customer and demand. A high level of supply and demand uncertainties results in lack of collaboration (Eftekhar et al. 2017) and influences the effective collaboration between public and private partners (Gabler, Richey, and Stewart 2017).

Measurement issues complicate tasks for collaboration, which aims to provide timely and appropriate responses. The lack of measurement systems prevents organisational learning (Maon, Lindgreen, and Vanhamme 2009). Furthermore, the most important measurements often change based on the characteristics of a disruption (Day 2014).

Resources of organisations are also considered as an important factor. Superior financial capabilities (Akhtar, Marr, and Garnevska 2012), the availability of transportation (Baharmand, Comes, and Lauras 2017), or human resources (Brusset and Teller 2017) are vital for any collaboration activities.

Visibility is also a factor in collaboration. The lack of visibility leads to information asymmetric (Dubey et al. 2018), especially for complex supply chain networks (Hosseini and Ivanov forthcoming). Thus, information technology that promotes visibility has been received much attention in the supply chain management area (Ergun et al. 2014).

Research synthesis

This section provides a research framework and identifies future research directions.

Research framework

Collaboration mechanisms were grouped into seven categories. It helps answer a critical question, in a specific disruption, which collaboration mechanism should be used? Figure 6 proposes the framework and some of the common collaborative mechanisms in each class.

Class 1 covers parametrical adaptation for small severe disruptions and focuses on two mechanisms: contractual and economics practices and joint practices. Adjustments on price and production quantity are useful in this class. Class 2 covers process adaptation and focuses on capacity flexibility. Technological and information sharing, joint practices, and relationship management are common mechanisms in this class. Training could increase the capabilities for responses. Sharing warehouse capacity could increase the utilisation and reduce costs of equipment and labour. Class 3 refers to structural adaption in the presence of severe disruptions. All collaboration mechanisms were found useful in this class. It is noted that research on commercial supply chain focus on parametrical adaptation while there are more research on humanitarian supply chain focus on structural adaptation. It is explained as there are many concerns relating to humanitarian issues when having severe disruptions.

Future research directions

The proposed research framework was developed through a holistic synthesis of the literature, identification of collaboration mechanisms, and influential factors in the response and recovery phases. Building on insights and research gaps, we present some future research directions as follows.

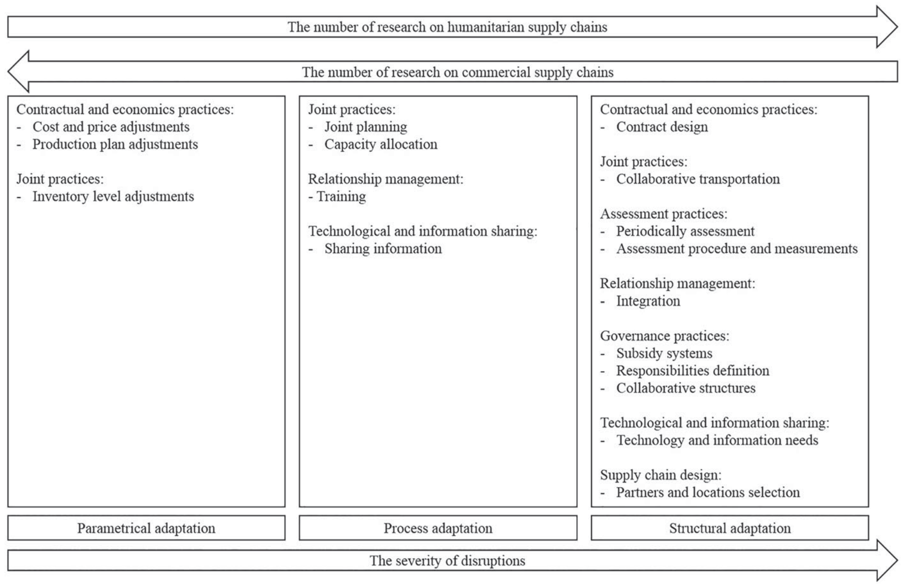

Figure 6. Supply chain collaboration framework for response and recovery.

Information sharing

Our results highlight the importance of information sharing in improving supply chain collaboration to respond and recover from disruptions. Information sharing increases visibility, velocity, and flexibility for the supply chain (Scholten and Schilder 2015). Once a disruption happens, supply chain partners need to share information to improve decision-making, to better match demand and supply, and to improve supply chains performance (Shen, Choi, and Minner 2019). Our findings show that there are contrary results on the role of information sharing in the presence of disruption. Sarkar and Kumar (2015) found that it is not beneficial for sharing disruption information at the downstream level. In contrast, Datta and Christopher (2011) showed that decentralised local stock information results in higher bullwhip and longer reaction time to disruptions. Additionally, lack of information or late receiving the right information reduce the flexibility to respond to disruptions as production schedules and stock receiving plans might be fixed already (Fayezi, Zutshi, and O'Loughlin 2017). We, therefore, call for studies that shed further lights on information exchange mechanisms.

A critical assumption in the extant literature is that the retailer shares information honestly (Yao, Xu, and Luan 2016). However, the attitude of possessing private information is common in the real world. Thus, information asymmetric exists and affects supply chain efficiency (Shen, Choi, and Minner 2019; Shi and Shen 2019). Therefore, future research should focus on information asymmetry and how to overcome or mitigate its negative effects. For example, in the presence of disruptions, a future study could investigate how to monitor the real demand by using big data techniques to track and trace information of upcoming orders and re-allocate these orders to match with the sudden demand. Alternatively, it is interesting to explore how to design supply chain contracts to overcome information asymmetric.

Competition

Our review shows that competitors have a critical role in reducing the effects of disruption. If retailer A does not have any available truck, there is always a competitor having a truck that retailer A could use. Although competition might cause overproduction from suppliers or over ordering from retailers (He and Zhao 2012), it increases flexibility and thus, resiliency for supply chains (Scholten and Schilder 2015). For example, when having retail competition, the retailer could collaborate and provide a loan to the manufacturer to avoid any disruption due to cash flow shortage (Shen et al. forthcoming). An appropriate supply chain contracts can help competing retailers and the manufacturer achieve a win-win outcome and survive in a highly uncertain market (Guo, Choi, and Shen 2020). As an example, Choi et al. (2019) show that for a given wholesale price contract, competing retailers could define optimal retail selling prices which optimise their expected profit. However, the rigorous analysis is still needed to explore the impact of competition on the supply chain collaboration. Future research can explore different types of supply chain contracts or the presence of information asymmetry in the competitive environment.

Performance measurement

Another key issue is to evaluate the damage after the disruption. However, not all commonly used measures are useful in evaluating and explaining the damage of disruptions (Altay and Ramirez 2010). This research issue has been highlighted strongly in humanitarian supply chains (Nurmala, de Vries, and de Leeuw 2018; Sigala and Wakolbinger 2019; Wagner and Thakur-Weigold 2018). These researchers have claimed that a set of performance measures is needed to assess the impact, select collaborative partners, and measure the success of a collaboration. Thus, we call for more research in this gap.

Incentives alignment

Motivating partners to collaborate and create values is another challenging issue. Incentive alignment, such as risk-sharing could enhance goal congruence and trust in supply chains (Eftekhar et al. 2017). Organisations could invest in maintaining and developing mutual trust for delivering a message of goal congruence in the presence of disruptions and reducing the possibility of false recognition in the collaboration process. Future research could investigate how to use incentive alignment to facilitate the process of contractual renegotiation and restore trust. Moreover, competition is desired and natural in supply chains. Thus, future research could investigate incentive mechanisms that engage competitors to increase supply chain resilience.

Costs

A common assumption in the literature of supply chain collaboration is that the cost involved in the collaboration is fixed or not significant. However, once a disruption happens, these costs will change and will affect the supply chain performance. For example, Wakolbinger and Cruz (2011) show that lower information-sharing costs may reduce total supply chain profit in some cases. Therefore, we need more studies that consider realistic costs and explore how these costs affect supply chain collaboration.

Leadership

Another issue in collaboration is the lack of research on the leadership for the collaboration. While most of the prior studies consider the leadership of the manufacturer, the role of the retailer should be studied more due to the growing power of retailers (Guo et al. 2017). It is also unclear who should lead the collaboration, especially under disruptive scenarios with multiple supply chain partners (Guo et al. 2017). These questions call for further research to yield convincing insights. For instance, when the demand is disrupted, a dominant retailer can offer incentives to stimulate demand and adjust the revenue sharing contract to avoid losses (Liu et al. 2017). However, it is unclear whether the leadership of the retailer is valid under other types of supply chain contracts. Thus, it is of great interest to investigate various leadership actors (e.g. manufacturers or retailers) in different collaboration mechanisms (e.g. different types of contract). Such understanding could provide insights on how to share risks, allocate resources, or identify common goals (Nurmala, de Vries, and de Leeuw 2018).

Methodological development

To pursue research directions identified in this research, we suggest a requirement for methodological development. The mathematical analysis or simulation methods are dominant in our sample as researchers aim at investigating different collaborative mechanisms. However, there are some efforts in using empirical research methods recently (e.g. Dubey et al. 2019). While we acknowledge the importance of mathematical analysis and simulation methods in investigating the collaboration mechanism, we encourage incorporating qualitative methods such as interviews or surveys with partners in the supply chain (Wagner and Thakur-Weigold 2018) to design and validate proposed models. Empirical studies such as case studies, field survey, or interviews are required to investigate fully the effects of disruption on supply chain collaboration and how partners collaborate to respond and recover from a disruption.

The success stories of collaboration after the disruptions could be observed from real practices. Among other factors, managing and understanding human might be the most challenging factors in supply chain collaboration (Kaur, Kanda, and Deshmukh 2008). Detailed empirical studies could disentangle complexities related to the human system and help to achieve collaboration. For example, while a supply chain contract could incur some costs to ensure the compliance of partners, these costs may destroy the long-term relationship (Harrison 2004) and the potential benefits of collaboration. Research should develop robust methodologies, which can evaluate the balance between drawback and benefits of supply chain contract. These types of studies help to provide full insights on how supply chain partners collaborate in the presence of disruptions.

Conclusion

Our systematic literature review identified, analysed, and synthesised literature to investigate how supply chain partners collaborate in the presence of disruption. Based on a sample of 157 papers, this research highlighted the complex processes that involve in the supply chain collaboration. Results from this review confirmed the usefulness of collaboration in response and recovery from disruptions for both commercial and humanitarian supply chains. Majority of these papers used mathematical modelling method, and since 2015 the survey method has been used frequently. We have identified seven common categories of collaborative mechanisms, including contractual and economics practices, joint practices, supply chain design, relationship management, assessment practices, technological and information sharing, and governance practices. Factors that influence collaboration include information sharing and technology, trust, culture, stakeholders, divergent goals, flexibility, knowledge and experience, market, measurement issues, resources, and visibility. Results of this review highlight research future directions relating to information sharing, competition, performance measurement, incentives alignment, costs, leadership, and methodology. For example, to fully understand the complexity, we advocate for the use of empirical studies that explore the contract mechanisms and information exchange systems. We noted that the results of this research are limited by the search engines and keywords. Therefore, a wider selection of keywords and search engines could broaden the results. Moreover, future research could consider commercial and humanitarian supply chains specifically to find practical ways to support the collaboration for each type of supply chains.

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